PROJECT REPORT · EHRSQL 2024 MIMIC-IV · JULY 2026

Governed Clinical Analytics Copilot

A self-hostable NL→SQL agent over de-identified EHR data, fine-tuned on OmniSQL-7B, achieving state-of-the-art performance on the EHRSQL 2024 benchmark with a fully open-source, 7-billion-parameter model.

OUR RS(10)	VS. EHRSQL 2024 WINNER	DELTA
0.873	**0.813**	**+7.4%**
Full test set · 1,167 questions	LG AI / KAIST · GPT-3.5 SFT + self-training	Open 7B model vs. proprietary GPT-3.5

Project Overview

Clinical data analysts spend hours writing SQL against complex EHR schemas. This project builds a governed, self-hostable copilot that translates plain-English questions into verified SQL queries over the MIMIC-IV database — and critically, **knows when to abstain** rather than hallucinate an answer for an unanswerable question.

We benchmarked against **EHRSQL 2024**, a shared task from NAACL 2024 ClinicalNLP challenging models with 1,167 test questions: 934 answerable and 233 intentionally unanswerable. The RS(10) metric penalises hallucinations 10× more than missed abstentions — making abstention quality the dominant factor.

Using **OmniSQL-7B** (Qwen2.5-Coder-7B-Instruct pre-trained on SynSQL-2.5M) as the base, we designed a supervised fine-tuning pipeline that teaches the model the MIMIC-IV schema, correct abstention behaviour, and zero-shot clinical NL→SQL — all without proprietary APIs.

Key Results

Metric	Value	Detail
RS(10) Score	**0.873**	Primary benchmark metric — full 1,167-question test set
Execution Accuracy	**92.4%**	863 / 934 answerable questions answered correctly
Hallucinations	**7 / 233**	Wrong SQL on unanswerable questions (3.0%)
Correct Abstentions	**226 / 233**	Model correctly refused 97% of unanswerable questions
Model Size	**7B params**	Open weights, self-hostable, no API dependency
Training Examples	**53,006**	SFT with 3× unanswerable oversampling
Training Hardware	**96 GB GPU**	Google Colab RTX 6000 Pro — 2 SFT epochs
Local Eval (100-Q)	**RS(10) = 0.940**	EX=92.5%, 0/20 hallucinations, 20/20 abstentions

Benchmark Comparison — EHRSQL 2024 MIMIC-IV

System	Model	EX	Halluc.	Abstain.	RS(10)
Ours (SFT-only) ★	OmniSQL-7B · open	92.4%	7/233	226/233	**0.873**
EHRSQL 2024 Winner (LG AI / KAIST)	GPT-3.5 SFT + self-training	~75%	—	—	0.813
EHRSQL 2024 3rd (ProbGate)	GPT-3.5 + log-prob filter	—	—	—	0.742
OmniSQL-7B baseline	No fine-tune	61.2%	133/233	100/233	–0.564

What We Built — End-to-End Pipeline

Phase 1: Data Engineering

Built 53,006 SFT training examples from EHRSQL train + valid + train_aug splits. Oversampled unanswerable questions 3× to match the 20% test distribution (vs. 8.9% in training data). Annotated answerable questions with gold SQL; unanswerable with [ABSTAIN]. Output: data/sft/omnisql_sft_train.jsonl (184 MB).

Phase 2: SFT Training — RTX 6000 Pro (96 GB)

Fine-tuned OmniSQL-7B with Unsloth + 4-bit quantisation. LoRA r=32, α=64, 7 target modules (q,k,v,o,gate,up,down). 2 epochs, effective batch size 16, cosine LR schedule at 1e-4. Trained on the full 2,733-character MIMIC-IV schema-inclusive system prompt. Adapter size: 323 MB, 392 LoRA tensors.

Phase 3: ORPO Preference Training

Built 492 preference pairs from baseline predictions (359 wrong-SQL + 133 hallucination pairs). Trained 1 additional epoch at lr=2e-6 with ORPO β=0.1. Key finding: SFT-only (0.873) outperformed SFT+ORPO due to asymmetric gradient dynamics — [ABSTAIN] is 1 token vs SQL at 50–200 tokens.

Phase 4: Evaluation Infrastructure

Built local evaluation harness (quick_eval_adapter.py) with stratified 80+20 sampling (SEED=42), batch inference, SQL execution against MIMIC-IV SQLite, and RS(N) scoring. Full harness supports self-consistency voting, entropy filtering, execution-guided repair, and RAG few-shot retrieval (BM25 + embed hybrid).

Critical Technical Lessons

System Prompt Must Include the Full Schema	The full prompt is 2,733 chars including the entire MIMIC-IV schema. The fallback stub (548 chars) provides no schema — model abstained on 99% of everything. A single file fix moved RS(10) from 0.210 → 0.873.
SFT Eval Format Must Match Training Format	Model trained on bare questions in the user turn. Few-shot eval prepended Q&A; preambles the model had never seen. Zero-shot (K=0) is the correct eval format for this model.

ORPO Overcorrected Abstention	[ABSTAIN] is 1 token (concentrated gradient) vs SQL at 50–200 tokens. With 27% abstention pairs, ORPO overcorrected — model abstained on 99%+ even when it should generate SQL. ORPO pairs must be built from the SFT model's own failures, not the baseline.
Unanswerable Class Imbalance Is the Core Challenge	Train: 8.9% unanswerable. Test: 20% unanswerable. Without 3× oversampling, the model never learns reliable abstention. Our strategy produced 97% correct abstention rate.
Entropy Calibration Must Match Eval Format	Post-SFT model entropy maxed at 0.0401 — effectively zero variation. Calibration must be re-run after any prompt or format change. With only 7 remaining hallucinations, correct calibration could push RS(10) → 0.90+.
Open 7B Beats Proprietary GPT-3.5	OmniSQL-7B's SynSQL-2.5M pre-training provides rich SQL patterns. Domain-specific SFT makes it exceed GPT-3.5 SFT + self-training at a fraction of the inference cost — fully on-premise with no API dependency.

What's Next

Next Step	Description	Potential Impact
Entropy Calibration	Re-calibrate abstention threshold with correct system prompt. Only 7 hallucinations remain.	**RS(10) → 0.90+**
Self-Training Pass	Apply SFT model to 38K train_aug examples, collect high-confidence [ABSTAIN] pseudo-negatives, run 1 additional	**RS(10) → 0.91–0.93**
Corrected ORPO Pairs	Build pairs from SFT model's own failures (not baseline). Cap abstention pairs at 1[illegible]5%. Use ORPO β=0.05.	**+1–2% RS(10)**
Governed Deployment	Package adapter + harness as self-hosted API with audit logging, query provenance, and role-based access control.	**Production milestone**